

- [SignedTemperature](#)
- [UnsignedTemperature](#)

While temperature values SHALL be transferred over the air using these types, this does not limit how Thermostats may display or store temperature values. Thermostats which display temperature values SHOULD follow the recommendations in [Conversion of Temperature Values for Display](#).

CAUTION

Calculations with temperature attributes

Where calculations or comparisons are performed, attribute values must be converted to a common type. In many cases, it is not sufficient to simply use the integer representation as the scaling from °C to integer value differs.

4.3.6. Setpoint Limits

There are a number of attributes which impose limits on setpoint values. This imposes constraints which SHALL be maintained by any mechanism which modifies a limit or setpoint. Individual attribute descriptions detail the actions to be taken should a conflict arise while modifying the value.

User configurable limits must be within device limits:

- [AbsMinHeatSetpointLimit](#) <= [MinHeatSetpointLimit](#) <= [MaxHeatSetpointLimit](#) <= [AbsMaxHeatSetpointLimit](#)
- [AbsMinCoolSetpointLimit](#) <= [MinCoolSetpointLimit](#) <= [MaxCoolSetpointLimit](#) <= [AbsMaxCoolSetpointLimit](#)

Setpoints must be within user configurable limits:

- [MinHeatSetpointLimit](#) <= [OccupiedHeatingSetpoint](#) <= [MaxHeatSetpointLimit](#)
- [MinCoolSetpointLimit](#) <= [OccupiedCoolingSetpoint](#) <= [MaxCoolSetpointLimit](#)
- [MinHeatSetpointLimit](#) <= [UnoccupiedHeatingSetpoint](#) <= [MaxHeatSetpointLimit](#)
- [MinCoolSetpointLimit](#) <= [UnoccupiedCoolingSetpoint](#) <= [MaxCoolSetpointLimit](#)

If, and only if, the AUTO feature is supported, a deadband must be maintained between Heating and Cooling setpoints and limits:

- [AbsMinHeatSetpointLimit](#) <= ([AbsMinCoolSetpointLimit](#) - [MinSetpointDeadBand](#))
- [AbsMaxHeatSetpointLimit](#) <= ([AbsMaxCoolSetpointLimit](#) - [MinSetpointDeadBand](#))
- [MinHeatSetpointLimit](#) <= ([MinCoolSetpointLimit](#) - [MinSetpointDeadBand](#))
- [MaxHeatSetpointLimit](#) <= ([MaxCoolSetpointLimit](#) - [MinSetpointDeadBand](#))
- [OccupiedHeatingSetpoint](#) <= ([OccupiedCoolingSetpoint](#) - [MinSetpointDeadBand](#))
- [UnoccupiedHeatingSetpoint](#) <= ([UnoccupiedCoolingSetpoint](#) - [MinSetpointDeadBand](#))